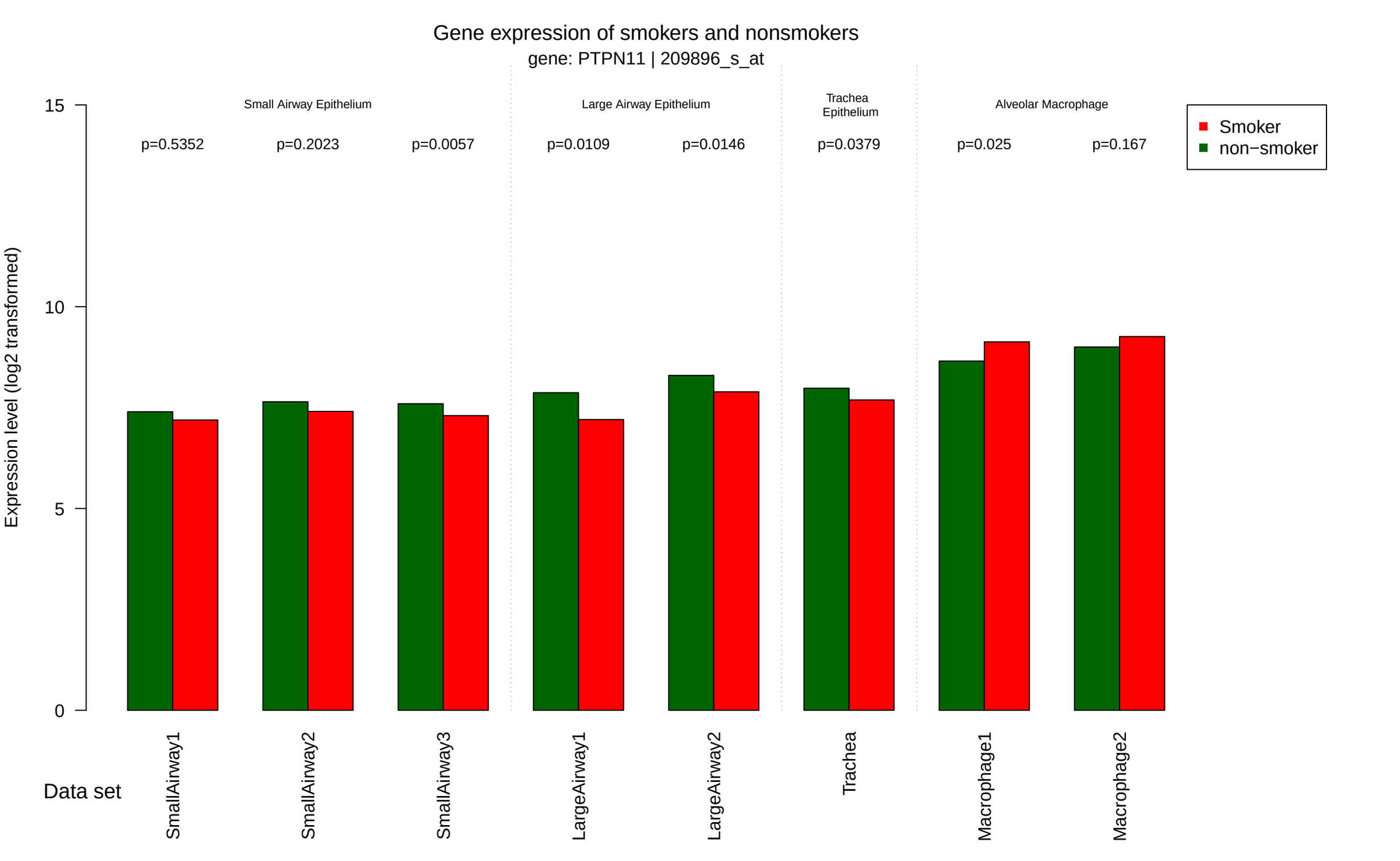
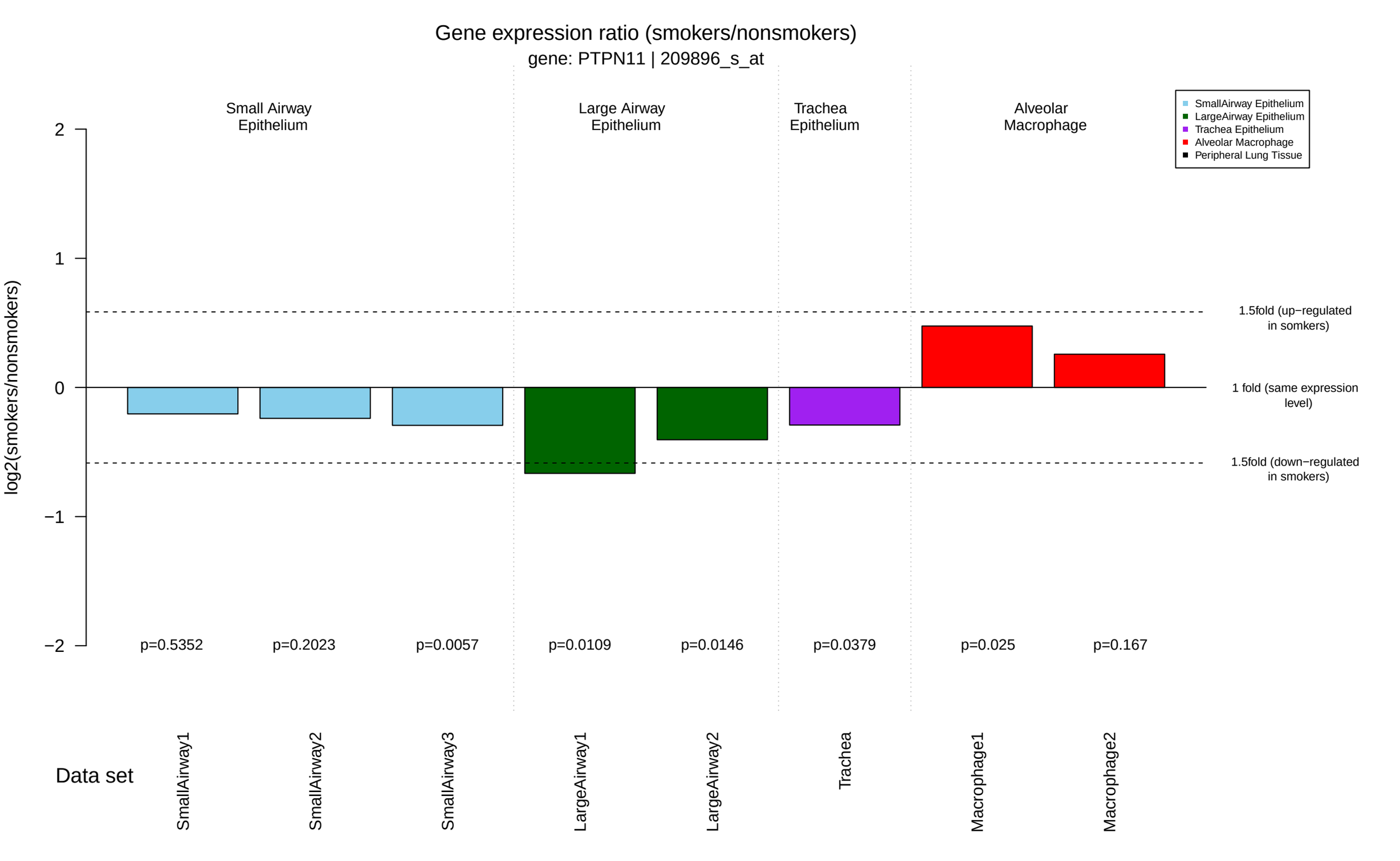
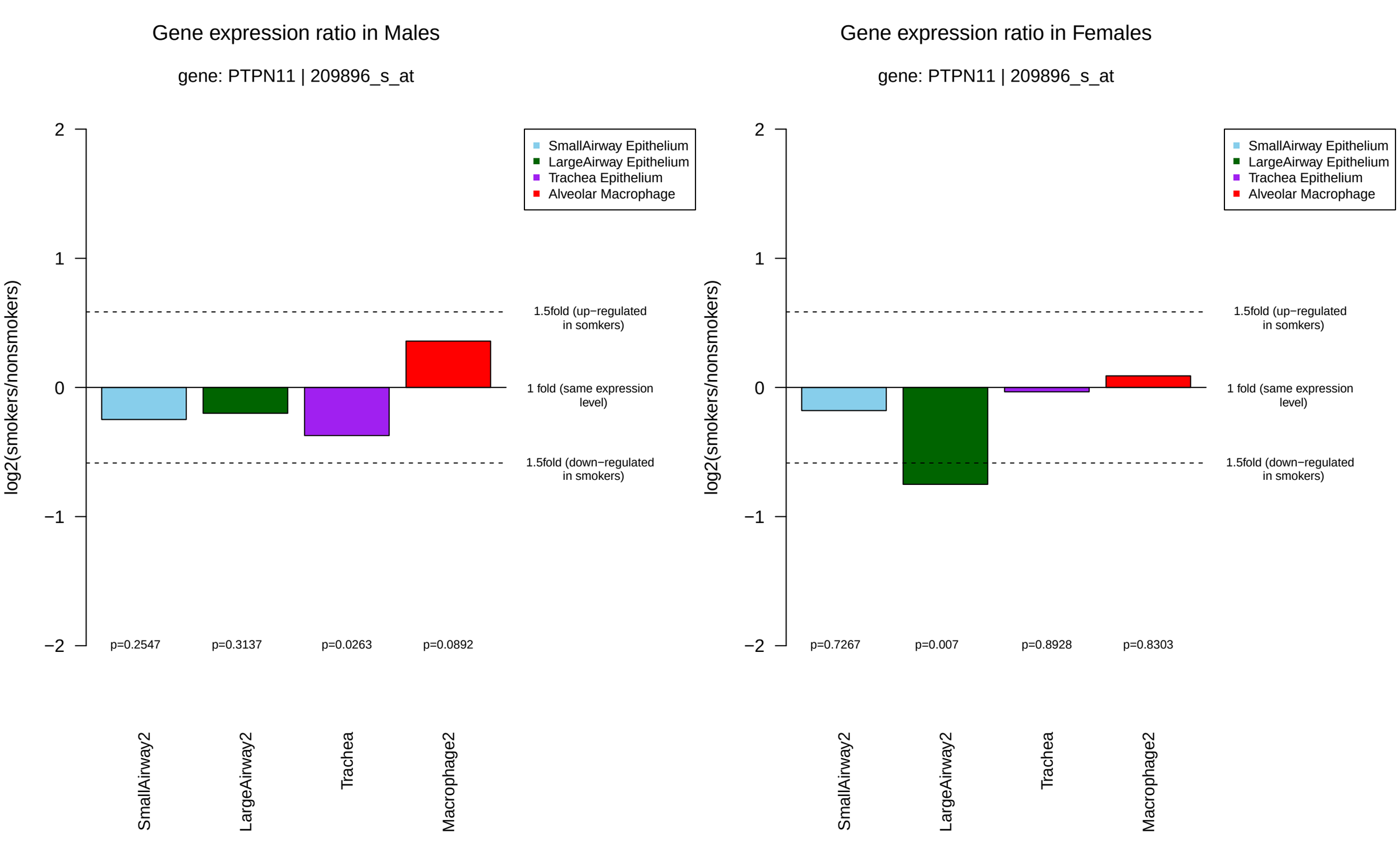
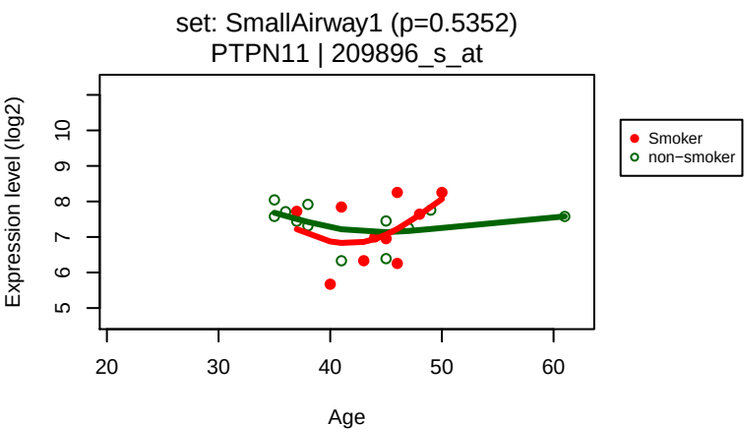
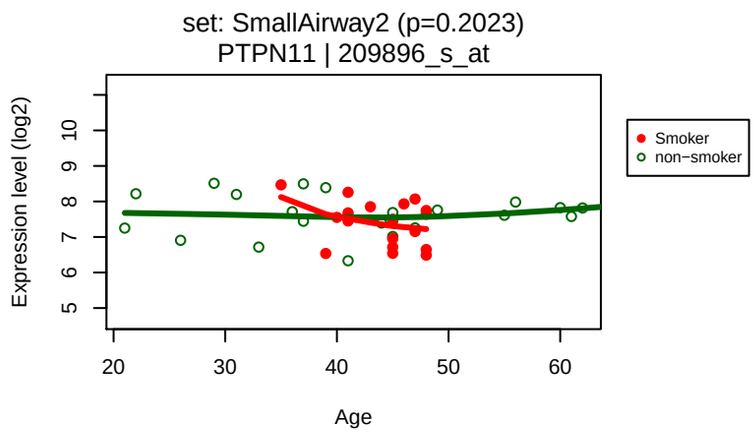
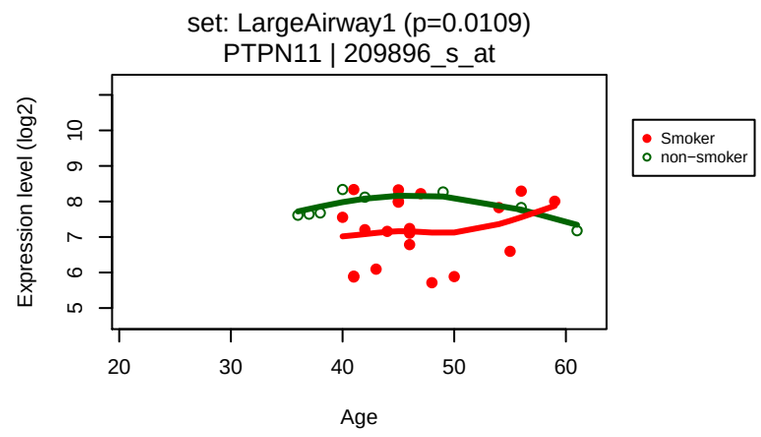

Gene expression of smokers and nonsmokers
gene: PTPN11 | 209896_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5352
p=0.2023
p=0.0057
p=0.0109
p=0.0146
p=0.0379
p=0.025
p=0.167
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

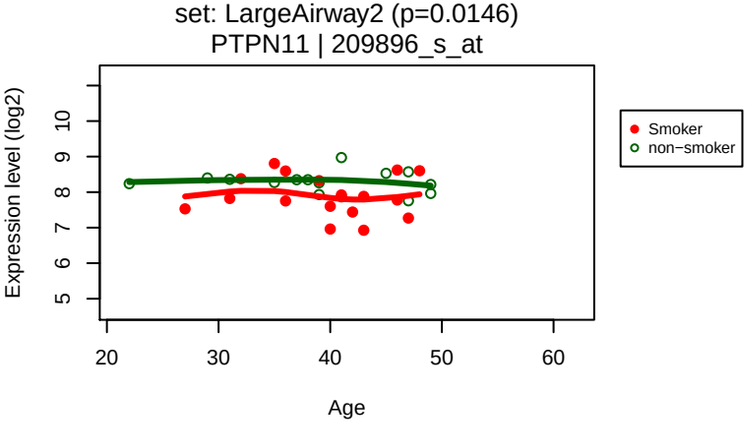

Gene expression ratio (smokers/nonsmokers)
gene: PTPN11 | 209896_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5352
p=0.2023
p=0.0057
p=0.0109
p=0.0146
p=0.0379
p=0.025
p=0.167
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

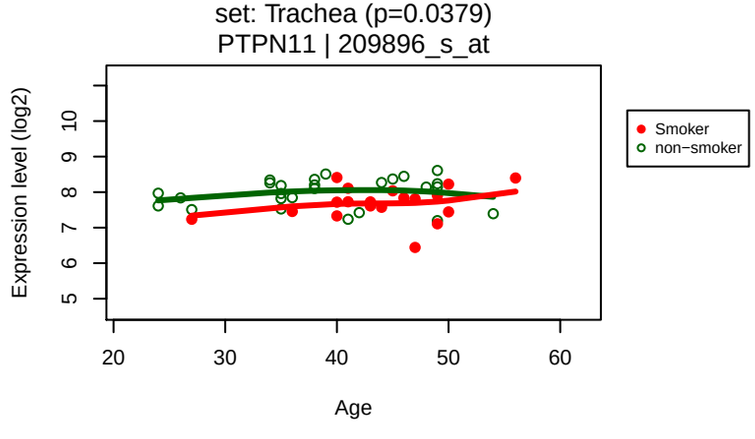

Gene expression ratio in Males
gene: PTPN11 | 209896_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2547
p=0.3137
p=0.0263
p=0.0892
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN11 | 209896_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7267
p=0.007
p=0.8928
p=0.8303
SmallAirway2
LargeAirway2
Trachea
Macrophage2

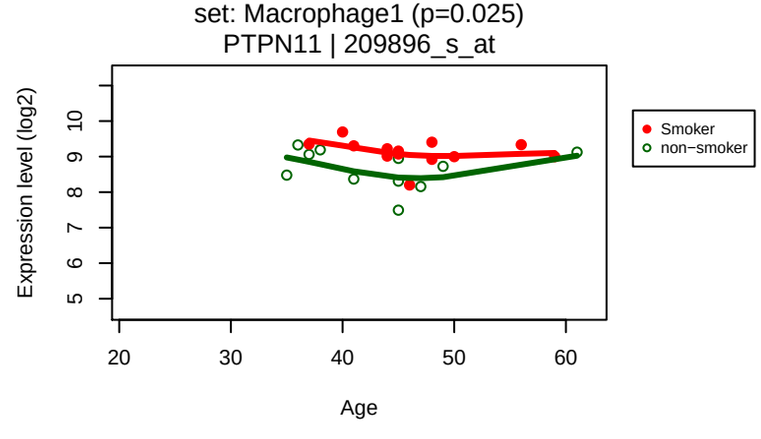

set: SmallAirway1 (p=0.5352)
PTPN11 | 209896_s_at
Expression level (log2)
Age
Smoker
non−smoker

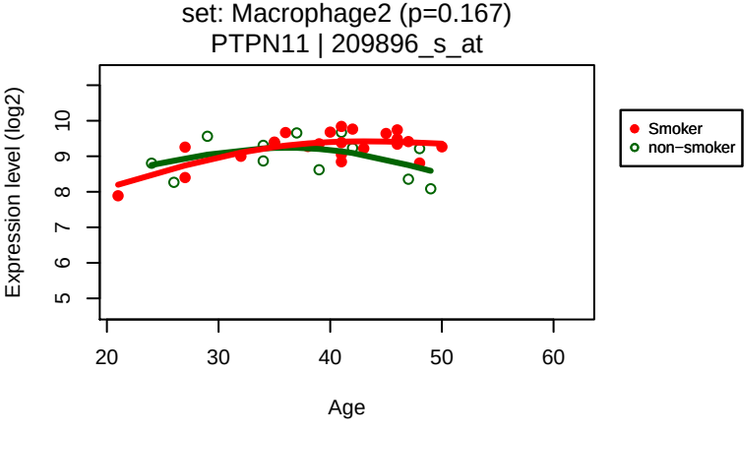

set: SmallAirway2 (p=0.2023)
PTPN11 | 209896_s_at
Expression level (log2)
Age
Smoker
non−smoker

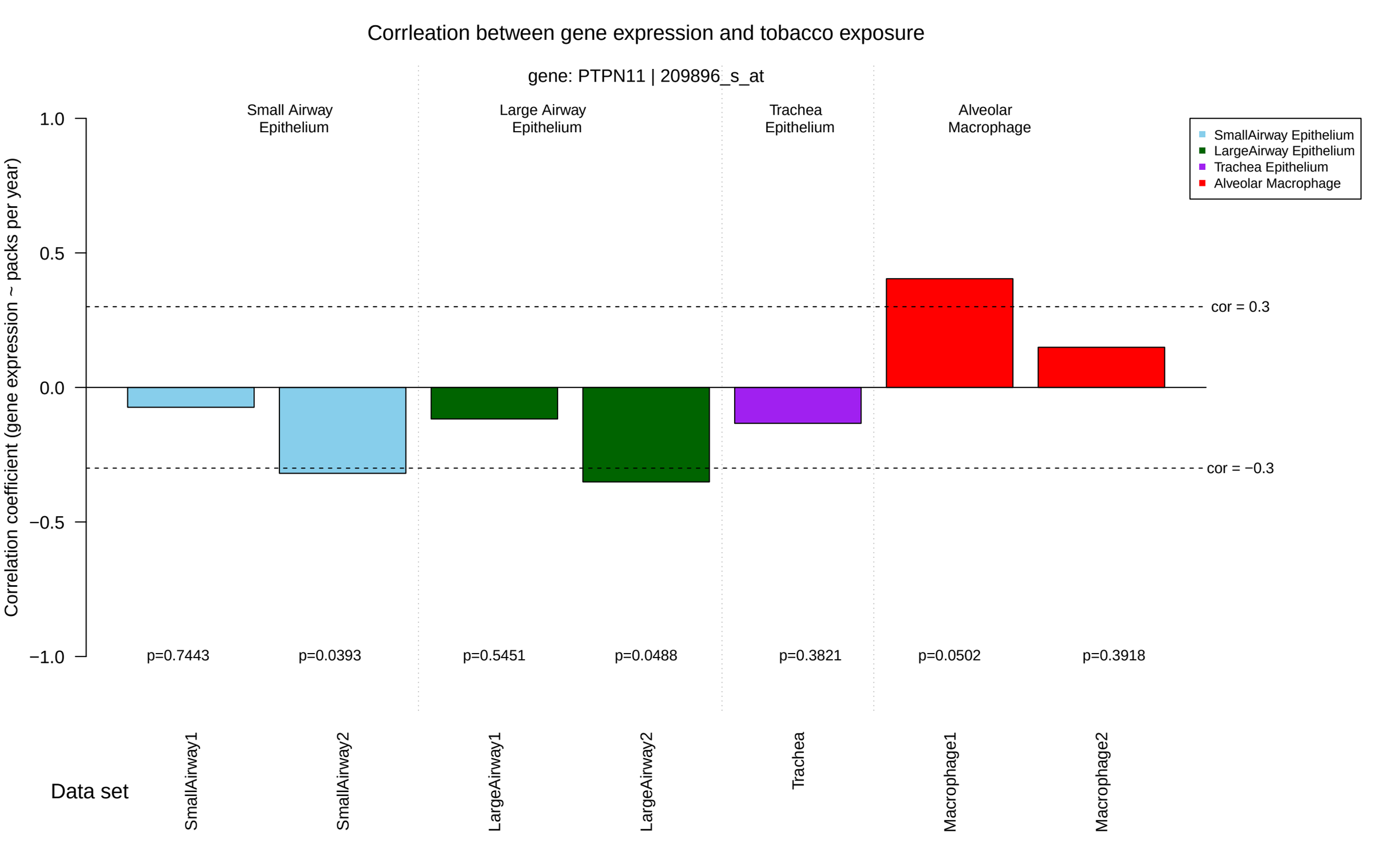

set: LargeAirway1 (p=0.0109)
PTPN11 | 209896_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0146)
PTPN11 | 209896_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0379)
PTPN11 | 209896_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.025)
PTPN11 | 209896_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.167)
PTPN11 | 209896_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN11 | 209896_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7443
p=0.0393
p=0.5451
p=0.0488
p=0.3821
p=0.0502
p=0.3918
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2